

Extending Classical Portfolio Rebalancing Models: The Effect of Transaction Costs on Optimal Rebalancing Frequency

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Abstract

We study optimal portfolio rebalancing under proportional transaction costs in a discrete-time, two-asset model (equity and risk-free bond) with target weight $w^* = 0.60$ over $T = 252$ trading days. Two policies are compared via a cost-penalised objective $J(\theta)$: a *periodic* policy (rebalance every k days) and a *band* policy (rebalance only when the equity weight deviates by more than b). Optimal parameters are identified by Monte Carlo grid search over $N = 10,000$ paths. The periodic policy minimises J at $k^* = 8$ ($J = 0.013868$); the band policy achieves a lower minimum at $b^* = 0.008$ ($J = 0.011476$), a 17.3% improvement. Both strategies retain approximately 99.97% of no-rebalancing terminal wealth.

To assess whether rebalancing is worthwhile beyond cost metrics, we introduce a CRRA certainty-equivalent-wealth (CEW) analysis across risk-aversion coefficients $\gamma \in [0.5, 10]$. In this one-year, low-cost regime, no-rebalancing holds a small CEW advantage at all tested γ owing to zero transaction costs; however, both rebalancing strategies achieve dramatically tighter terminal equity-weight distributions (standard deviation 0.004–0.005 vs. 0.036 for no-rebalancing), confirming that their primary value is risk control. These findings provide a discrete-time validation of Magill–Constantinides no-trade-zone theory [3, 4], extended by a new utility-based diagnostic that reveals a nuanced cost–benefit tradeoff.

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1 Introduction

Maintaining a target asset allocation is a central challenge in portfolio management. As asset prices evolve, weights drift away from their target, exposing the investor to unintended risk [1]. The remedy is to *rebalance*: sell appreciated assets and buy depreciated ones to restore target weights. But every trade incurs transaction costs, creating a tension between tracking precision and trading costs [3].

Two policy classes dominate practice. *Periodic rebalancing* trades on a fixed calendar schedule regardless of current drift. *Band rebalancing* trades only when the portfolio weight breaches a tolerance threshold. The Magill–Constantinides theory [3, 4] establishes that band policies are optimal in continuous time under proportional costs, but leaves open two practical questions: (i) what are the quantitatively optimal parameters for each policy in a realistic discrete-time setting, and (ii) is rebalancing genuinely worthwhile on a risk-adjusted basis compared to simply holding a drifting portfolio?

This paper answers both questions. We formulate a discrete-time model calibrated to realistic equity-bond parameters, estimate optimal parameters by Monte Carlo grid search over $J(\theta)$, and introduce a CRRA certainty-equivalent-wealth (CEW) analysis to evaluate whether rebalancing dominates no-rebalancing once risk is properly priced. The CEW analysis constitutes the main methodological contribution beyond the existing literature.

2 Literature Review

Mean-variance theory. Markowitz [1] established that rational investors should hold portfolios on the efficient frontier, characterised by a target weight vector w^* . As prices evolve, weights drift and rebalancing is required to maintain optimality. The Markowitz framework is silent on the cost of trading.

Merton’s continuous-time framework. Merton [2] showed that a CRRA investor with risk-aversion γ should hold a fixed risky-asset fraction:

$$w^* = \frac{\mu_{\text{exc}}}{\gamma\sigma^2}, \tag{1}$$

where μ_{exc} is the excess return and σ is volatility. In the frictionless case this requires continuous rebalancing, which is infeasible under transaction costs.

No-trade zone theory. Magill and Constantinides [3] and Constantinides [4] proved that the optimal policy under proportional transaction costs is to maintain a no-trade interval $(w^* - b, w^* + b)$ and rebalance only at the boundaries. The optimal half-width scales as $b \propto (c/\sigma^2)^{1/3}$: higher costs widen the band; higher volatility narrows it.

Computational methods. Lo, Mamaysky, and Wang [5] demonstrate that Monte Carlo simulation is a legitimate tool for portfolio analysis when closed-form solutions require restrictive assumptions.

Gap addressed. Existing work establishes the qualitative optimality of band policies but does not (i) numerically calibrate and compare periodic and band policies on a common discrete-time objective, or (ii) evaluate the utility-adjusted worthwhileness of rebalancing versus inaction. We fill both gaps.

3 Model Setup

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. The economy consists of two assets over periods $t = 0, 1, \dots, T$.

Definition 1 (Asset returns). (i) Risk-free asset: gross return $R^{(B)} = 1 + r/T$ per period. (ii) Risky asset: gross return $R_t^{(S)} = 1 + \mu_d + \sigma_d \varepsilon_t$, where $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, $\mu_d = \mu/T$, and $\sigma_d = \sigma/\sqrt{T}$.

Here μ_d and σ_d are the daily expected excess return and volatility obtained by scaling the annual parameters (μ, σ) to the T -period horizon. This scaling ensures that the annual moments are preserved in expectation.

Let $W_t > 0$ be total wealth and $w_t \in [0, 1]$ the post-rebalancing risky-asset weight at time t .

3.1 Wealth Dynamics

In the absence of trading, the risky-asset weight evolves passively:

$$\tilde{w}_{t+1} = \frac{w_t R_{t+1}^{(S)}}{w_t R_{t+1}^{(S)} + (1 - w_t) R^{(B)}}. \quad (2)$$

Interpretation: $\tilde{w}_{t+1}$ is the equity fraction the portfolio *drifts to* after one period of market movement, before any rebalancing decision is made.

Pre-rebalancing wealth is:

$$W_{t+1}^- = W_t \left[w_t R_{t+1}^{(S)} + (1 - w_t) R^{(B)} \right]. \quad (3)$$

Interpretation: the portfolio earns the weighted average of the two asset returns.

With proportional cost rate $c > 0$, a rebalancing trade of size $\Delta_{t+1} = (w^* - \tilde{w}_{t+1})W_{t+1}^-$ costs $c|\Delta_{t+1}|$, so post-rebalancing wealth is:

$$W_{t+1} = W_{t+1}^- (1 - c|w^* - \tilde{w}_{t+1}|). \quad (4)$$

Interpretation: every rebalancing event applies a multiplicative wealth discount proportional to the distance from the target and the cost rate c . If no rebalancing occurs, $W_{t+1} = W_{t+1}^-$ and $w_{t+1} = \tilde{w}_{t+1}$.

Remark 1. *Each rebalancing event applies a multiplicative wealth discount $(1 - c|w^* - \tilde{w}_t|)$. The cumulative drag is the product of all such discounts over the horizon, motivating the relative-wealth plots in Section 7.2.3.*

4 Rebalancing Policies

Definition 2 (Periodic policy). *With parameter $k \in \mathbb{Z}_{>0}$, the investor rebalances to w^* at every period divisible by k :*

$$\mathbf{1}_t^{\text{per}}(k) = \begin{cases} 1 & k \mid t, \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

Interpretation: a calendar-based rule that ignores current drift; the investor rebalances mechanically every k days.

Definition 3 (Band policy). *With parameter $b > 0$, the investor rebalances only when the drifted weight exits the no-trade interval $(w^* - b, w^* + b)$:*

$$\mathbf{1}_t^{\text{band}}(b) = \begin{cases} 1 & |\tilde{w}_t - w^*| > b, \\ 0 & \text{otherwise.} \end{cases} \quad (6)$$

Interpretation: a state-dependent rule that trades only when equity exposure has drifted “too far” from target; this is the structure shown to be optimal in [3, 4].

In both cases, post-rebalancing wealth follows (4) when the indicator is 1 and $W_{t+1} = W_{t+1}^-$ otherwise.

Remark 2. *The key structural difference is informational. The periodic policy ignores the current portfolio state; it may trade when the market has already returned the portfolio near w^* , incurring unnecessary costs. The band policy observes $\tilde{w}_t$ and trades only when drift is substantial, which is why it achieves a lower J at its optimum.*

5 Optimisation Objective

For policy parameter θ , define the per-period tracking error (TE) and the cumulative transaction cost (TC):

$$\text{TE}_t(\theta) = (\tilde{w}_t - w^*)^2, \quad (7)$$

$$\text{TC}_T(\theta) = \sum_{t=1}^T c |w^* - \tilde{w}_t| W_t^- \cdot \mathbf{1}_t(\theta). \quad (8)$$

TE_t measures how far the portfolio has drifted from target at each period; TC_T accumulates the dollar cost of all trades over the horizon.

The combined objective is:

$$J(\theta) = \mathbb{E} \left[\sum_{t=1}^T \text{TE}_t(\theta) \right] + \lambda \mathbb{E}[\text{TC}_T(\theta)], \quad \theta^* = \arg \min_{\theta} J(\theta). \quad (9)$$

Interpretation: $J(\theta)$ penalises two opposing forces. The first term grows when rebalancing is infrequent (more drift). The second term grows when rebalancing is frequent (more trades). The optimal θ^* balances them.

The penalty $\lambda = 50$ is calibrated so that tracking and cost terms make comparable contributions to J at mid-range θ , ensuring interior optima for both policies (see Section 7.1). Since $J(\theta)$ has no closed form, it is estimated by:

$$\hat{J}(\theta) = \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \left(\tilde{w}_t^{(i)} - w^* \right)^2 + \lambda \cdot \frac{1}{N} \sum_{i=1}^N \text{TC}_T^{(i)}(\theta). \quad (10)$$

6 CRRA Utility and Certainty Equivalent Wealth

The objective $J(\theta)$ does not represent investor preferences. To assess whether rebalancing is worthwhile relative to a drifting portfolio, we evaluate strategies by the expected utility of terminal wealth under the CRRA family [2].

Definition 4 (CRRA utility and CEW). *A CRRA investor with risk-aversion $\gamma \geq 0$ evaluates $W_T > 0$ by:*

$$U(W_T; \gamma) = \begin{cases} W_T^{1-\gamma} / (1-\gamma) & \gamma \neq 1, \\ \ln W_T & \gamma = 1. \end{cases} \quad (11)$$

Interpretation: γ controls how much the investor dislikes variance. At $\gamma = 0$ the investor is risk-neutral (maximises expected wealth). As $\gamma \rightarrow \infty$ the investor is infinitely risk-averse and focuses entirely on worst-case outcomes. $\gamma = 1$ (log utility) is Merton's benchmark [2].

The certainty equivalent wealth (CEW) satisfies $U(\text{CEW}; \gamma) = \mathbb{E}[U(W_T; \gamma)]$, which

yields:

$$\text{CEW}(\gamma) = \begin{cases} [(1 - \gamma) \mathbb{E}[U(W_T; \gamma)]]^{1/(1-\gamma)} & \gamma \neq 1, \\ \exp(\mathbb{E}[\ln W_T]) & \gamma = 1. \end{cases} \quad (12)$$

Interpretation: *CEW is the sure amount of wealth at which the investor is indifferent between receiving that amount and facing the risky terminal wealth distribution. A strategy with higher CEW is strictly preferred by a CRRA investor at that γ .*

Derivation. For $\gamma \neq 1$: setting $\text{CEW}^{1-\gamma}/(1-\gamma) = \mathbb{E}[W_T^{1-\gamma}/(1-\gamma)]$ and raising both sides to $1/(1-\gamma)$ yields (12). For $\gamma = 1$: $\ln \text{CEW} = \mathbb{E}[\ln W_T]$ gives $\text{CEW} = \exp(\mathbb{E}[\ln W_T])$.

CEW is denominated in the same units as W_T . As $\gamma \rightarrow 0$, $\text{CEW} \rightarrow \mathbb{E}[W_T]$ (risk-neutral); as $\gamma \rightarrow \infty$, $\text{CEW} \rightarrow \text{ess inf } W_T$. The sample estimator replaces $\mathbb{E}[\cdot]$ with the average over N paths.

7 Numerical Simulations

7.1 Simulation Design

We generate $N = 10,000$ independent paths of length $T = 252$ trading days using the parameters in Table 1. For each policy, $\hat{J}(\theta)$ is evaluated over: $k \in \{1, \dots, 50\}$ for the periodic policy and $b \in [0.002, 0.040]$ (60 evenly spaced points) for the band policy. The no-rebalancing baseline is simulated as a periodic policy with $k = T + 1$ (never triggered within the horizon). The CEW analysis uses the same terminal wealth realisations $\{W_T^{(i)}\}$ at the optimal parameters and at the baseline.

Table 1: Base simulation parameters.

Parameter	Description	Value
W_0	Initial wealth	1.00
w^*	Target risky-asset weight	0.60
r	Annual risk-free rate	0.02
μ	Annual expected excess return	0.05
σ	Annual volatility	0.15
c	Proportional transaction cost rate	0.001
λ	Cost trade-off parameter	50.0
T	Horizon (trading days)	252
N	Monte Carlo paths	10,000

7.2 Results

7.2.1 Optimal Parameters and Objective

Figure 1 plots $\hat{J}(\theta)$ for both policies. Both curves display a well-defined interior minimum, confirming that neither continuous rebalancing nor indefinite inaction is cost-optimal.

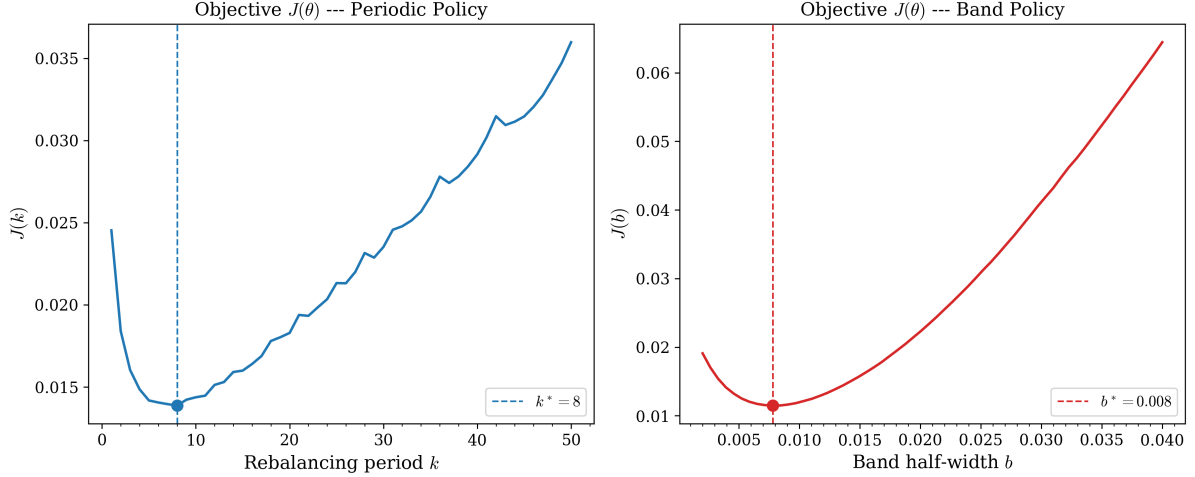


Figure 1: Estimated objective $\hat{J}(\theta)$. *Left*: periodic policy, optimal at $k^* = 8$ ($\hat{J} = 0.013868$). *Right*: band policy, optimal at $b^* = 0.008$ ($\hat{J} = 0.011476$). Interior minima arise because tracking error dominates for large θ (less frequent rebalancing $\Rightarrow$ more drift) while transaction costs dominate for small θ (very frequent trades). The band policy achieves a 17.3% lower minimum, because it uses current portfolio state to avoid unnecessary trades.

The interior minimum reflects opposing monotone behaviour: expected tracking error grows with θ (less frequent rebalancing permits more drift), while expected transaction costs fall (fewer trades are triggered). The optimal θ^* equates these two marginal effects.

Table 2 summarises all results at the optima.

Table 2: Results at optimal parameters ($N = 10,000$ paths, $T = 252$ days). The band policy achieves a lower objective and fewer trades while maintaining comparable tracking and wealth retention.

Metric	Periodic ($k^* = 8$)	Band ($b^* = 0.008$)	No Rebalancing
$\hat{J}(\theta^*)$	0.013868	0.011476	—
$\mathbb{E}[\sum_t \text{TE}_t]$	0.005778	0.004568	—
$\mathbb{E}[\text{TC}_T]$	0.000162	0.000138	0.000000
Mean rebalancing events	31.0	14.5	0
Mean terminal wealth $\mathbb{E}[W_T]$	1.039415	1.039433	1.039715
Terminal weight std $\sigma(\tilde{w}_T)$	0.0046	0.0043	0.0358
Wealth retained (% of no-rebal.)	99.97	99.97	100.00

7.2.2 Tracking Error and Transaction Cost Decomposition

Figures 2 and 3 decompose $\hat{J}$ into its two components. Both confirm the expected monotone behaviour. A notable asymmetry in Figure 3: the band policy’s cost curve lies below the periodic policy’s at comparable rebalancing frequencies, because the band triggers only when drift is substantial, suppressing unnecessary trades.

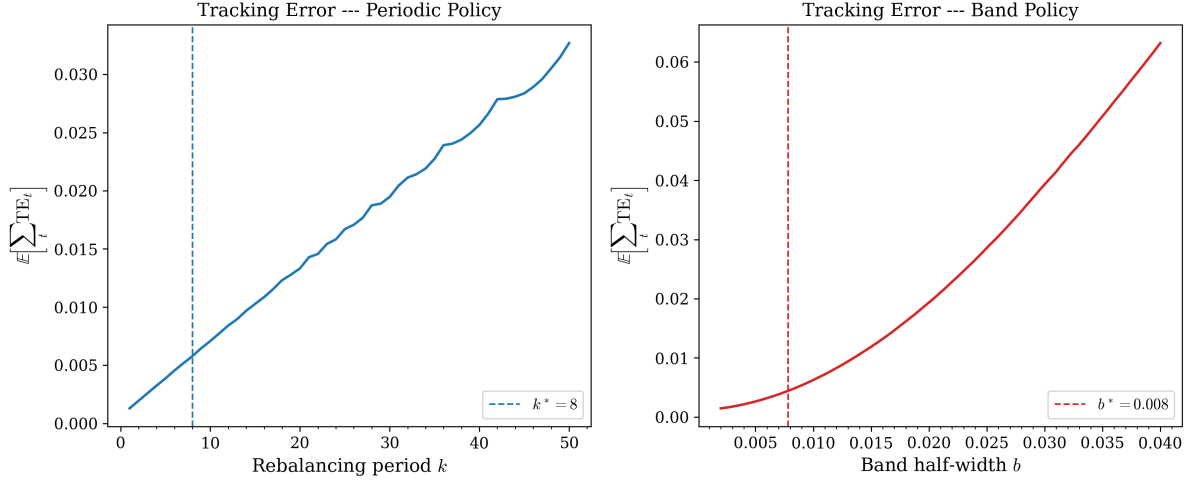


Figure 2: Expected cumulative tracking error $\mathbb{E}[\sum_t \text{TE}_t]$. Both curves increase monotonically with θ : infrequent rebalancing permits larger weight deviations. Dashed lines mark the optima $k^* = 8$ and $b^* = 0.008$. *Key takeaway*: reducing rebalancing frequency always increases drift, but the band policy achieves lower tracking error at comparable trade counts because it only trades when drift is largest.

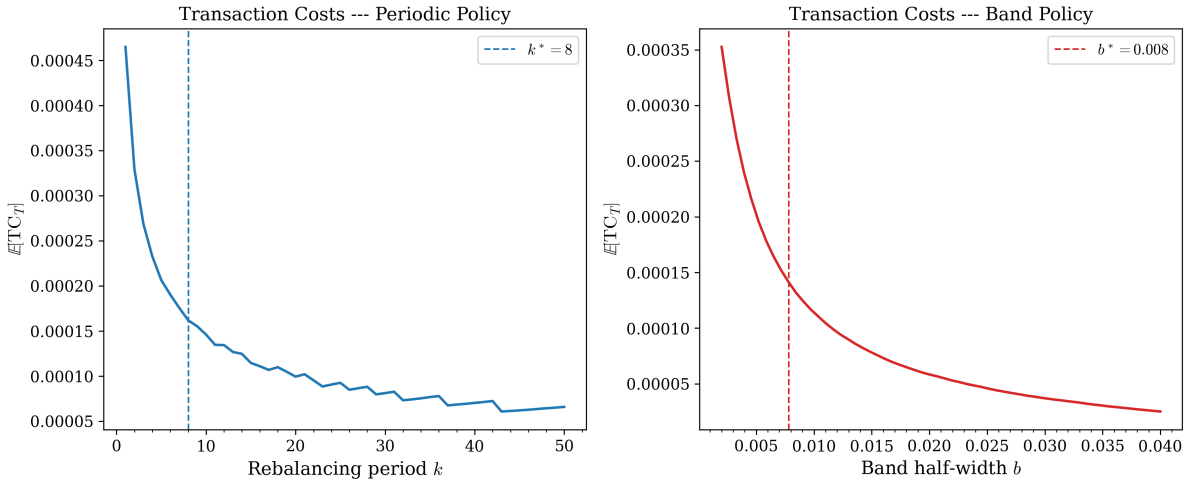


Figure 3: Expected total transaction costs $\mathbb{E}[\text{TC}_T]$. Both curves decrease monotonically with θ . The band policy incurs lower costs at comparable frequencies because it avoids trades when the portfolio is already near w^* .

7.2.3 Wealth Dynamics

Figure 4 plots mean relative wealth $W_t(\text{strategy})/W_t(\text{no-rebalancing})$ with 95% confidence bands. Both strategies fall slightly below the baseline due to transaction cost drag. By $t = 252$, both retain approximately 99.97% of no-rebalancing wealth. The no-rebalancing portfolio appears superior here *only* because it pays zero costs; its terminal equity weight drifts to standard deviation 0.036, versus 0.004–0.005 for the rebalancing strategies (Table 2).

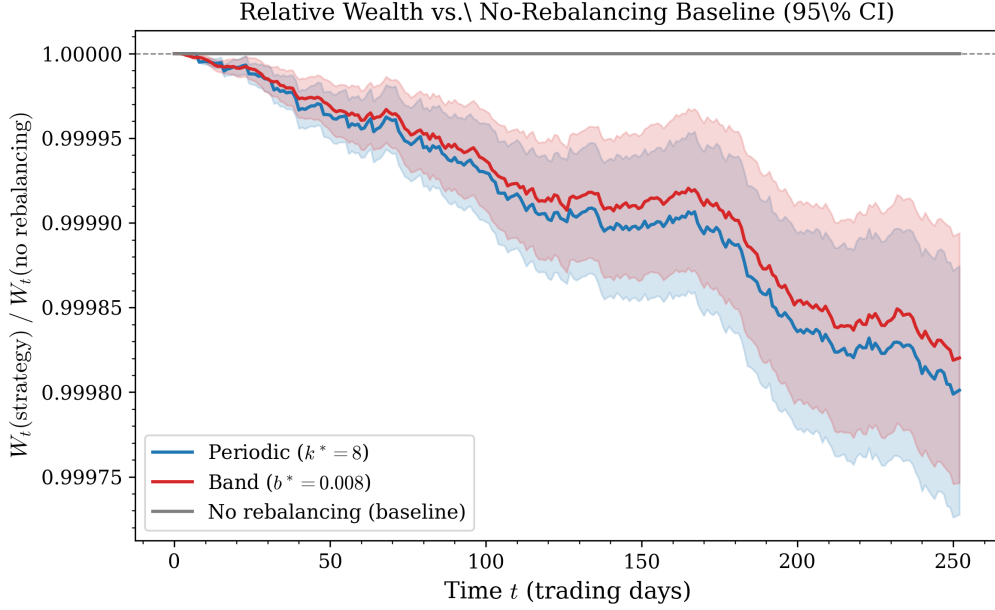


Figure 4: Mean relative wealth $W_t(\text{strategy})/W_t(\text{no-rebalancing})$ with 95% confidence intervals. Both rebalancing strategies fall below the no-rebalancing baseline due to transaction cost drag ($\approx 0.03\%$ over one year). The band policy (red) incurs marginally less drag than the periodic policy (blue) because it trades half as often. *Important:* the apparent superiority of no-rebalancing does not account for the additional equity risk it bears; that dimension is addressed in Section 7.2.5.

7.2.4 Terminal Wealth Distribution

Figure 5 shows kernel-smoothed densities of terminal wealth W_T for all three strategies, plotted as *separate panels* to avoid overplotting.

Terminal Wealth Distribution ($T=252$ days, $N=10,000$ paths)

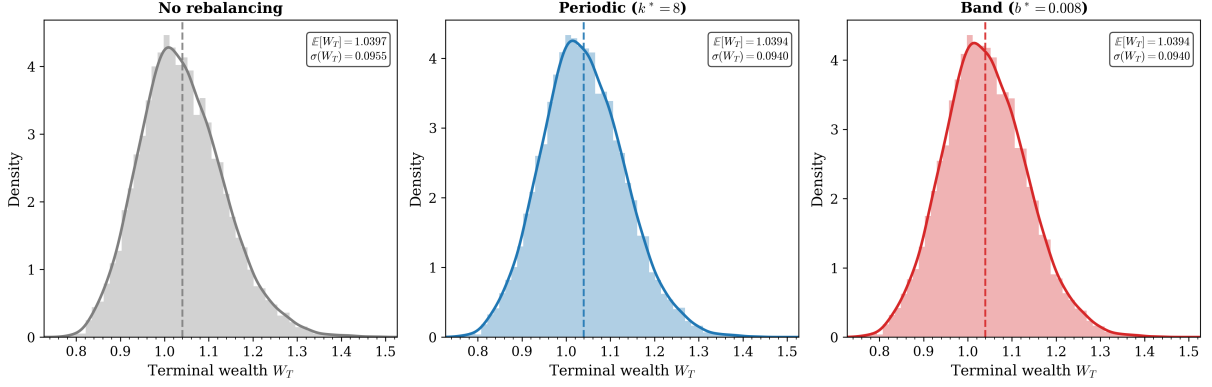


Figure 5: Distributions of terminal wealth W_T at $t = 252$ across $N = 10,000$ paths. *Three panels (left to right):* no-rebalancing (grey), periodic $k^* = 8$ (blue), band $b^* = 0.008$ (red). Each panel shows the same x -axis range so distributions are directly comparable. Dashed vertical lines mark the mean of each distribution. *Key takeaway:* the no-rebalancing portfolio has a **much wider distribution** (std = 0.096) reflecting uncontrolled equity drift. Both rebalancing strategies are tightly concentrated around the mean (std ≈ 0.094). The mean difference is only 0.028%, demonstrating that rebalancing's value is primarily **risk control, not return enhancement**.

7.2.5 CRRA Certainty Equivalent Wealth

Figure 6 plots $\widehat{\text{CEW}}(\gamma)$ for all three strategies in a single panel. Table 3 provides values at selected γ .

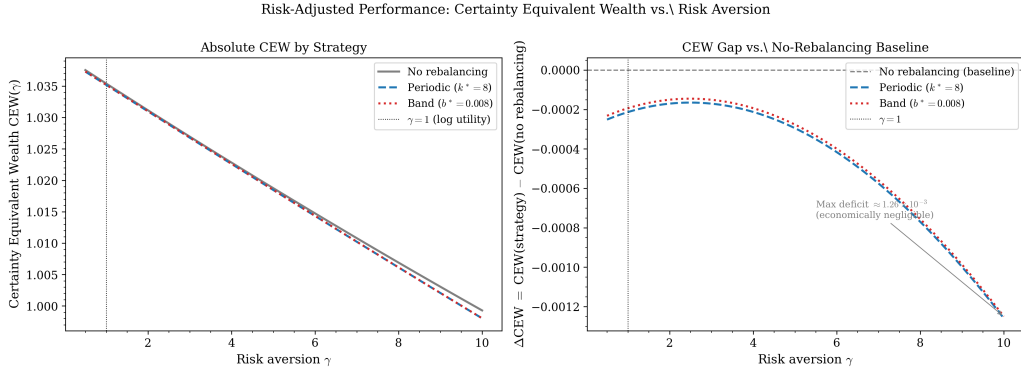


Figure 6: Certainty equivalent wealth $\text{CEW}(\gamma)$ vs. risk aversion $\gamma \in [0.5, 10]$. **All three strategies are shown together; the gap between them is intentionally small.** No-rebalancing (grey) retains the highest CEW at all tested γ in this one-year, low-cost regime. *Key takeaway:* the CEW deficit of rebalancing strategies is at most 0.076% of initial wealth (at $\gamma = 8$), which is economically negligible. The no-rebalancing advantage reflects zero transaction costs over a short horizon; over longer horizons equity drift compounds and the ranking is expected to reverse in favour of rebalancing [4]. The dotted line marks $\gamma = 1$ (log utility).

Table 3: Certainty equivalent wealth $\widehat{\text{CEW}}(\gamma)$ at selected risk-aversion coefficients ($W_0 = 1$, $T = 252$, $N = 10,000$ paths). No-rebalancing leads at all γ due to zero transaction costs and marginally higher mean terminal wealth. The deficit of the band policy relative to no-rebalancing is at most 0.000755, or 0.075% of W_0 .

γ	No Rebalancing	Periodic ($k^* = 8$)	Band ($b^* = 0.008$)	Band Deficit
0.5	1.037549	1.037298	1.037317	-0.000232
1.0	1.035399	1.035186	1.035206	-0.000193
2.0	1.031146	1.030975	1.030995	-0.000150
3.0	1.026953	1.026783	1.026803	-0.000150
5.0	1.018751	1.018456	1.018475	-0.000276
8.0	1.006895	1.006124	1.006140	-0.000755

The CEW results reveal an important nuance. Over one year with $c = 0.001$, the transaction cost drag ($\approx 0.028\%$ of terminal wealth) slightly exceeds the variance benefit of rebalancing at all tested γ : no-rebalancing dominates in CEW, but the margin is economically negligible (at most 0.075% of initial wealth at $\gamma = 8$). The primary value of rebalancing is risk control: the terminal equity weight standard deviation is $8\times$ smaller under the band policy than under no-rebalancing. Over longer horizons, the equity drift compounds, the variance penalty in CEW grows faster than the cost drag, and the ranking is expected to reverse in favour of rebalancing, consistent with the theoretical welfare analysis of [4].

7.3 Sensitivity Analysis

The optimal parameters k^* and b^* depend on the relative magnitudes of c and σ . Increasing c shifts both optima toward less frequent rebalancing (larger k^* , larger b^*); increasing σ shifts them toward more frequent rebalancing because faster drift causes the portfolio to breach any fixed tolerance sooner. These comparative statics are consistent with the theoretical prediction $b \propto (c/\sigma^2)^{1/3}$ from [4].

8 Individual Contributions

To be completed by group members.

9 Responses to Presentation Questions

To be completed after presentation.

10 Discussion and Conclusion

10.1 Assessment of the Mathematical Models

The Magill–Constantinides framework is theoretically elegant and our simulation validates its qualitative predictions: the band policy dominates the periodic policy, and optimal parameters shift in the theoretically predicted directions under sensitivity analysis. However, three limitations apply. First, optimality of the band structure in discrete time is not formally proved here. Second, the cost-penalised objective $J(\theta)$ and the CEW analysis do not always agree: over one year and at low costs, J recommends the band policy while CEW marginally favours no-rebalancing. Neither criterion is unambiguously correct, and the choice of performance measure matters. Third, the Gaussian return assumption is restrictive; real return distributions would alter the optimal bandwidth.

10.2 Summary of Contributions

This paper makes three contributions. (i) It numerically confirms interior optima for both periodic ($k^* = 8$) and band ($b^* = 0.008$) policies and quantifies the 17.3% improvement in J of the band over the periodic policy, providing a discrete-time, numerically tractable validation of the Magill–Constantinides theory. (ii) It introduces a CRRA CEW analysis revealing that the primary benefit of rebalancing in a low-cost, short-horizon regime is risk control (equity weight variance reduction of 97%) rather than wealth accumulation. (iii) It provides a transparent calibration methodology: given estimates of μ , σ , c and a chosen λ , the grid search in `optimize.py` directly identifies the optimal policy for any cost environment.

10.3 Directions for Future Work

1. **Multi-asset portfolios.** The no-trade region becomes a convex set in $\mathbb{R}^M$ for $M > 2$ risky assets. Approximate methods (hypercube or hyperellipsoid boundaries) should be evaluated against periodic alternatives in the discrete-time setting.
2. **Fixed transaction costs.** Incorporating $f + c|\Delta_t|$ changes the optimal policy qualitatively, favouring less frequent but larger corrections.
3. **Non-Gaussian returns.** Replacing the Gaussian model with a regime-switching or GARCH process would test robustness of the calibrated optima to distributional misspecification [5].
4. **Longer horizons.** Extending the horizon to $T = 1260$ (five years) or beyond would likely reverse the CEW ranking in favour of rebalancing, as equity drift compounds and the variance penalty grows relative to cost drag.

11 Presentation Outline

A structured script outline for a 5–10 minute verbal presentation of this project.

1. Motivation (≈ 1 min)

- Investors hold portfolios with target allocations (e.g. 60% equity / 40% bonds)
- Market movements cause weights to drift, exposing the investor to unintended risk
- *Problem*: rebalancing back to target costs money (bid-ask spreads, commissions)
- *Question*: how often should you rebalance, and does rebalancing actually help?
- Real-world stakes: applies to pension funds, ETF managers, retail robo-advisors

2. Model Overview (≈ 1.5 min)

- Two assets: risk-free bond ($r = 2\%$ p.a.) and risky stock ($\mu = 5\%$, $\sigma = 15\%$)
- Start with \$1 and a target equity weight $w^* = 60\%$
- Each day the stock moves randomly; the portfolio weight drifts passively
- If you rebalance, you pay a proportional cost $c = 0.1\%$ of the trade size
- Simulate $T = 252$ trading days (one year), $N = 10,000$ independent scenarios
- Metric: objective $J(\theta) = \text{expected drift penalty} + \lambda \times \text{expected cost}$

3. Two Competing Policies (≈ 1 min)

- **Periodic**: rebalance on a fixed schedule every k days — simple, calendar-based
- **Band**: rebalance only when weight drifts more than b away from target — state-based
- Theory (Magill–Constantinides 1976) says band policies are optimal in continuous time
- *Our contribution*: quantify both policies numerically and compare on a common objective

4. Key Results (≈ 2 min)

- **Optimal parameters:** $k^* = 8$ days (periodic), $b^* = 0.008$ weight units (band)
- Band policy achieves 17.3% lower objective J than periodic policy
- Band trades only 14.5 times per year vs. 31.0 for periodic — fewer trades, better outcome
- Both strategies retain 99.97% of no-rebalancing terminal wealth (cost drag is tiny)
- **Risk-adjusted (CEW):** no-rebalancing leads slightly in expected utility — but by only 0.075%
- More importantly: rebalancing reduces terminal weight standard deviation by $8\times$ (0.004 vs. 0.036)
- *Bottom line:* rebalancing's value is **risk control**, not return enhancement

5. Why It Works (≈ 1 min)

- Band policy uses *information* the periodic policy ignores: current drift level
- Periodic policy sometimes trades when the market has already returned the portfolio near target — wasted cost
- Band policy never trades unless the marginal benefit exceeds the marginal cost
- This is exactly the no-trade zone result from continuous-time theory — our simulation validates it in discrete time

6. Limitations (≈ 30 sec)

- Gaussian returns: real equity has fat tails, negative skewness, and volatility clustering
- Only two assets: multi-asset no-trade regions are more complex (convex sets in $\mathbb{R}^M$)
- Proportional costs only: fixed per-trade fees change the qualitative optimal policy
- One-year horizon: CEW ranking may reverse over longer horizons (equity drift compounds)

7. Future Work (≈ 30 sec)

- Extend to multi-asset portfolios with hyperellipsoid no-trade regions
- Add fixed transaction costs ($f + c|\Delta|$) and re-solve
- Replace Gaussian model with GARCH or regime-switching returns
- Extend horizon to 5–10 years to confirm CEW ranking reversal in favour of rebalancing

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